FLUTTER MOBILE APP

PROJECT PLAN SUBMISSION

ComfyGo

A mobile tourist management system for discovering and booking travel services across Bangladesh.

PLATFORM

Flutter and Dart
Android and iOS

PROPOSED FINAL MILESTONE

22 November 2026

This plan defines a realistic three-phase path for building the ComfyGo mobile application with navigation, validated forms, Provider state management, REST API data handling, reusable widgets, and testing.

SUBMITTED BY

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Project Plan

ComfyGo mobile tourist management system

Project Title	ComfyGo	**Platform**	Flutter for Android and iOS
Category	Tourism and travel management	**State Management**	Provider
Submitted By	Partho Deb	**Student ID**	241-134-016
SWE Batch	6	**Plan Deadline**	13 September 2026

SECTION 01

Project Title

ComfyGo is a Flutter-based mobile tourist management application for travel within Bangladesh.

SECTION 02

Project Description

ComfyGo will bring the main services of the existing web platform into a mobile application. Users will be able to create an account, discover tourist spots, browse hotels, tours, packages, transport services, and tour guides, and manage bookings from one interface. The application will be developed with Flutter and Dart so that one codebase can support both Android and iOS.

The semester version will focus on a complete tourist journey and basic role-based dashboards. Flutter widgets will provide the interface, Provider will manage shared application state, REST API services will exchange JSON data with the existing FastAPI backend, and SharedPreferences will store small local settings such as the theme and login session. This scope is large enough to demonstrate course concepts but limited enough to complete in three phases.

SECTION 03

Problem Statement

Domestic travellers often use separate sources to find destinations, compare hotels, locate transport, contact guides, and track bookings. Information may be inconsistent, so trip planning takes more time. Service providers also need a simple way to view requests and keep availability information current.

ComfyGo will organize these services in one mobile application with search, detail screens, validated forms, role-based access, and booking and payment-status tracking. The semester version will record payment choices and statuses through the backend. Live financial gateway integration will remain optional unless time and course approval allow it.

SECTION 04

Target Users

ComfyGo is designed for the following users:

- Domestic tourists and visitors who want to discover destinations and arrange travel services from a phone.
- Hotel managers who need a basic mobile view of room availability and booking requests.
- Tour guides who need to maintain their profile and availability and review hire requests.
- Administrators who monitor users, listings, bookings, payments, membership tiers, and contact messages.

SECTION 05

List of Core Features

The required features prioritize the tourist booking flow. Basic provider and administrator functions are included so the application still reflects the multi-role design of ComfyGo.

Feature	What the mobile application will provide
Authentication and Role Access	Registration, login, logout, session handling, and role-based navigation for tourists, hotel managers, tour guides, and administrators.
Home and Tourism Discovery	A mobile home screen with categories and lists for tourist spots, hotels, tours, packages, guides, and transport services.
Search and Filtering	Search and filter available services by useful fields such as location, category, price range, and availability.
Tourist Spot Details	Destination descriptions and essential travel information presented through reusable cards and detail screens.

Feature	What the mobile application will provide
Hotel Booking	Hotel and room details, availability display, validated booking forms, and booking confirmation or status.
Tours and Packages	Browse tours and travel packages, review price and schedule details, and create a booking request.
Transport Services	Browse transport options such as bus, train, air, or launch where supported by the backend, then submit a booking request.
Tour Guide Hiring	View guide profiles, languages, areas, ratings, and availability, then submit a guide-hire request.
Booking and Payment Status	A unified booking history with pending, confirmed, completed, failed, or cancelled states and recorded payment information.
Account and Contact	Profile management, membership or subscription information, and a contact form for questions or support.
Basic Role Dashboards	Focused views for managers, guides, and administrators to review relevant requests, availability, and summary information.

SECTION 06

Technical Approach and Course Concept Coverage

The mobile application will use a simple layered structure so that screens, state, and data-access code remain separate. This approach makes individual parts easier to test and update during the semester.

Course Concept	Application in ComfyGo
Multi Screen Navigation	Splash, onboarding, login, register, home, list, details, booking, history, profile, and role dashboard screens connected with named routes and bottom navigation.
Forms and Validation	Registration, login, booking, profile, availability, and contact forms with required-field, format, date, quantity, and range validation.
Provider State Management	AuthProvider, CatalogProvider, BookingProvider, and ThemeProvider will expose shared state and notify the UI when data changes.
REST API and JSON	Service classes will call the existing FastAPI endpoints, parse JSON into Dart models, and handle loading, success, empty, and error states.
Local Persistence	SharedPreferences will store the login token or session flag, theme preference, onboarding completion, and lightweight settings.
Reusable Widgets	Examples include ServiceCard, SpotCard, HotelCard, GuideCard, StatusChip, EmptyState, LoadingView, and validated input components.
Lists and Responsive Layout	ListView, GridView, MediaQuery, and LayoutBuilder will support different phone sizes and scrollable service catalogues.
Testing	Unit tests for models, validation, providers, and service logic; widget tests for important forms and screens; integration tests for the main booking flow.

SECTION 07

Development Milestones With Deadlines

The project plan is due on 13 September 2026. The phase dates below are proposed working deadlines for progress tracking and may be adjusted after instructor feedback.

Phase	Dates and Deadline	Planned Work and Measurable Output
Phase 1 **Foundation and UI**	20 September to 4 October 2026 Deadline 4 October	Set up the Flutter project and Git repository; define folders, theme, routes, and reusable widgets; create the main screen structures with mock data. Output: a navigable UI prototype.
Phase 2 **Core Features and Data**	5 October to 1 November 2026 Deadline 1 November	Create Dart models, API services, and Provider classes; connect authentication and catalogue data; implement validated forms, local settings, search, booking flows, status handling, and basic role dashboards. Output: a functional app connected to the backend.
Phase 3 **Testing and Finalization**	2 November to 22 November 2026 Deadline 22 November	Test providers, services, forms, widgets, and the booking flow; fix defects; refine error states, responsiveness, and accessibility; complete the README, user guide, presentation, and release build. Output: a stable documented app ready for demonstration.

SECTION 08

Expected Outcome

By the end of Phase 3, ComfyGo should provide a stable mobile workflow from account access and service discovery to booking submission and status review. The completed project should demonstrate Flutter navigation, reusable widgets, validated forms, Provider state management, asynchronous API communication, local preferences, responsive design, testing, and clear documentation.

The priority is to complete and test the required booking journey before adding optional improvements. Features that depend on external paid services, live payment gateways, or advanced map integration will be treated as stretch goals so they do not delay the core semester submission.